

Chapter 1

Sets and Categories

1.1 Sets

1.1.1 Sets and Functions

sets, functions

1.1.2 The Axiom of Choice

Zorn's lemma, WOP, transfinite induction

1.1.3 Cantor-Schröder-Bernstein Theorem

proving CSB

1.2 Categories

1.2.1 Categories

Definition 1.2.1. A *category* $\mathcal{C}$ consists of:

- A collection of objects $\text{ob}(\mathcal{C})$
- For each $A, B \in \mathcal{C}$, a set $\text{Hom}_{\mathcal{C}}(A, B)$ of morphisms from A to B
- An associative composition map, where the composition of $f \in \text{Hom}_{\mathcal{C}}(A, B)$ and $g \in \text{Hom}_{\mathcal{C}}(B, C)$ is denoted $g \circ f \in \text{Hom}_{\mathcal{C}}(A, C)$
- For each $A \in \text{ob } \mathcal{C}$, there is an identity map $\text{id}_A \in \text{Hom}_{\mathcal{C}}(A, A)$ such that $f \circ \text{id}_A = f$ for all $f \in \text{Hom}_{\mathcal{C}}(A, B)$ and $\text{id}_A \circ g = g$ for all $g \in \text{Hom}_{\mathcal{C}}(B, A)$.

Remark 1.2.2. Some people only require that morphisms only form a collection, not a set. Categories where the collections of morphisms do form a set are called *locally small*. Categories where the collection of objects forms a set are called *small*.

Example. The category **Set** of all sets has ob Set as the collection of all sets, and $\text{Hom}_{\text{Set}}(A, B)$ as the set of set-functions $A \rightarrow B$.

Example. The category **Div** has $\text{ob Div} = \mathbb{N} = \{1, 2, \dots\}$ and $\text{Hom}_{\text{Div}}(a, b) = \begin{cases} \{*\} & a \mid b \\ \emptyset & a \nmid b \end{cases}$, where $*$ denotes some object (the details are not specific). If $a \mid b$ and $b \mid c$, then $a \mid c$, and the composition of the morphisms $a \rightarrow b$ and $b \rightarrow c$ is given in the only way possible: $* \circ * = *$.

Example. For any poset P , there is a category $\mathcal{C}$ with $\text{ob } \mathcal{C} = P$ and $\text{Hom}_{\mathcal{C}}(A, B) = \begin{cases} \{*\} & A \leq B \\ \emptyset & \text{else} \end{cases}$. More generally, if $R \subseteq S \times S$ is a reflexive and transitive relation, we can form a category $\mathcal{C}$ with $\text{ob } \mathcal{C} = S$ and $\text{Hom}_{\mathcal{C}}(a, b) = \begin{cases} \{*\} & aRb \\ \emptyset & \text{else} \end{cases}$.

Example. For any category $\mathcal{C}$, there is an *opposite category* $\mathcal{C}^{\text{op}}$ given by “reversing all the arrows”. That is, $\text{ob } \mathcal{C}^{\text{op}} = \text{ob } \mathcal{C}$ and $\text{Hom}_{\mathcal{C}^{\text{op}}}(A, B) = \text{Hom}_{\mathcal{C}}(B, A)$ with $f \circ_{\mathcal{C}^{\text{op}}} g = g \circ f$.

We’ll write $f : A \rightarrow B$ to mean $f \in \text{Hom}_{\mathcal{C}}(A, B)$. We’ll say that $g : B \rightarrow A$ is a left inverse to f if $g \circ f = \text{id}_A$, and a right inverse to f if $f \circ g = \text{id}_B$. We say g is a two-sided inverse of f if it is both a left and right inverse of f . In the case that f has a left inverse g and a right inverse h , then $g = g \circ \text{id}_B = g \circ f \circ h = \text{id}_A \circ h = h$, and $g = h$ is a two-sided inverse to f .

Definition 1.2.3. A morphism $f : A \rightarrow B$ is said to be:

- *Epic* (or an *epimorphism*) if $g_1 \circ f = g_2 \circ f \implies g_1 = g_2$ for all $g_1, g_2 : B \rightarrow C$
- *Monic* (or a *monomorphism*) if $f \circ g_1 = f \circ g_2 \implies g_1 = g_2$ for all $g_1, g_2 : C \rightarrow A$
- An *isomorphism* if it has a two-sided inverse

Example. In the category **Set**, epimorphisms are precisely surjections, monomorphisms are precisely injections, and isomorphisms are precisely bijections.

The composition of two epimorphisms is again an epimorphism, and the composition of two monomorphisms is again a monomorphism. Epimorphisms become monomorphisms in the opposite category and vice versa.

In general, isomorphisms are both epimorphisms and monomorphisms, but the converse is not true (for example, consider the map $[0, 1] \rightarrow S^1$ given by $x \mapsto e^{2\pi i x}$ in the category of topological spaces). This is because having a right (resp. left) inverse is sufficient to be an epimorphism (resp. monomorphism), but the condition is not necessary.

If f is an isomorphism, its inverse is unique (if g and h are both inverses, then g is in particular a left inverse and h is in particular a right inverse, so we must have $g = h$). We can thus write f^{-1} for the inverse of an isomorphism f . Thus, the composition $g \circ f$ of two isomorphisms f and g is an isomorphism with inverse $f^{-1} \circ g^{-1}$. It also therefore makes sense to consider categories where all maps are isomorphisms. Such categories are called *groupoids*. For instance, categories arising from equivalence relations are groupoids.

Definition 1.2.4. An *endomorphism* of A is a morphism $A \rightarrow A$. We write $\text{End}_{\mathcal{C}}(A) = \text{Hom}_{\mathcal{C}}(A, A)$. An *automorphism* of A is an isomorphism $A \rightarrow A$, and we write $\text{Aut}_{\mathcal{C}}(A)$ for the set of all automorphisms of A .

1.2.2 Constructing Categories (Cones)

Given a category $\mathcal{C}$, we can construct multiple other categories from it. If $A \in \mathcal{C}$, the slice category $\mathcal{C}_A$ has objects $(Z, f : Z \rightarrow A)$ and morphisms $g : Z_1 \rightarrow Z_2$ such that the diagram

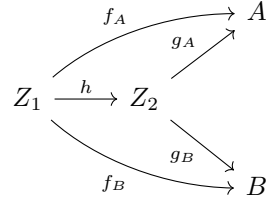
$$\begin{array}{ccc} Z_1 & \xrightarrow{g} & Z_2 \\ & \searrow f_1 & \swarrow f_2 \\ & A & \end{array}$$

commutes. The identity map of (Z, f) is id_Z . Similarly, the coslice category $\mathcal{C}^A$ has objects $(Z, f : A \rightarrow Z)$.

Example. If $I \in \text{ob } \mathcal{C}$ is an initial object, $\mathcal{C}_I$ adds no extra information, so is in a sense “the same” as the category $\mathcal{C}$.

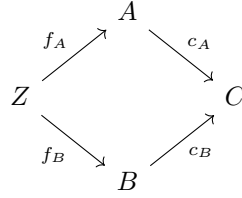
Example. In the case that $\mathcal{C} = \mathbf{Set}$ and $A = \{*\}$, the coslice category, denoted $\mathbf{Set}^*$ consists of pairs (S, s) , where $s \in S$ is a distinguished element. The morphisms $(S, s) \rightarrow (T, t)$ are set maps $f : S \rightarrow T$ with $f(s) = t$. This is the category of pointed sets.

For $A, B \in \text{ob } \mathcal{C}$, we can construct the more complicated category $\mathcal{C}_{A,B}$ with objects $(Z, f_A : Z \rightarrow A, f_B : Z \rightarrow B)$. Morphisms $(Z_1, f_A, f_B) \rightarrow (Z_2, g_A, g_B)$ are morphisms $h : Z_1 \rightarrow Z_2$ such that

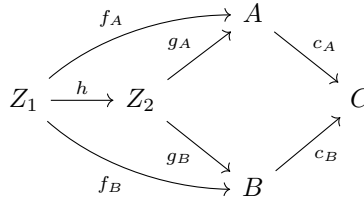


commutes. Once again, there is similarly the category $\mathcal{C}^{A,B}$ with objects $(Z, f_A : A \rightarrow Z, f_B : B \rightarrow Z)$.

We can construct a fibered version of the category $\mathcal{C}_{A,B}$. Given $c_A : A \rightarrow C$ and $c_B : B \rightarrow C$, the objects are $(Z, f_A : Z \rightarrow A, f_B : Z \rightarrow B)$ such that

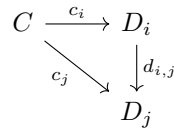


commutes. Morphisms $(Z_1, f_A, f_B) \rightarrow (Z_2, g_A, g_B)$ are morphisms $h : Z_1 \rightarrow Z_2$ such that

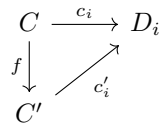


commutes.

More generally, given a “diagram” D (which we can think of as a collection of objects together with morphisms between those objects), we can form the category $\text{cone}(D)$ whose objects are $C \in \text{ob } \mathcal{C}$ together with maps $f_i : C \rightarrow D_i$ for each object D_i in D , such that



commutes for every morphism $d_{i,j} : D_i \rightarrow D_j$ in D . We'll succulently represent this as a pair $(C, c : C \rightarrow D)$. Morphisms between $(C, c : C \rightarrow D)$ and $(C', c' : C' \rightarrow D)$ are morphisms $f : C \rightarrow C'$ such that



commutes for each object D_i in D . Similarly, the category $\text{cocone}(D)$ is defined as pairs $(C, c : D \rightarrow C)$.

1.2.3 Universality

The use of category theory is that it provides a language for encapsulating “universal properties”, which are canonical ways of describing objects.

Definition 1.2.5. An object $I \in \text{ob } \mathcal{C}$ is said to be *initial* if, for all $A \in \text{ob } \mathcal{C}$, $\text{Hom}_{\mathcal{C}}(I, A)$ has exactly one element. An object $F \in \text{ob } \mathcal{C}$ is said to be *final* if, for all $A \in \text{ob } \mathcal{C}$, $\text{Hom}_{\mathcal{C}}(A, F)$ has exactly one element. We say $Z \in \text{ob } \mathcal{C}$ is a *zero object* if it is both initial and final.

Initial and final objects need not exist in an arbitrary category.

Example. In the category **Set**, $\emptyset$ is initial and $\{*\}$ is final, where $\{*\}$ denotes any singleton set. In the category **Div**, 1 is initial and there is no final object. In the category **Set**^{*}, $(\{*\}, *)$ is both initial and final, and thus a zero object.

Example. In the category $\mathcal{C}_A$, (A, id_A) is final. If $I \in \text{ob } \mathcal{C}$ is initial and $f : I \rightarrow A$, then (I, f) is initial.

Initial objects become final objects in the opposite category and vice versa. In particular, zero objects stay zero objects.

Proposition 1.2.6. *Initial (resp. final) objects are unique up to (unique) isomorphism.*

Proof. Since final objects are initial in the opposite category, it suffices to show this for initial objects. If I and I' are initial, there are unique $f : I \rightarrow I'$ and $g : I' \rightarrow I$. Now, $f \circ g : I' \rightarrow I'$ and $g \circ f : I \rightarrow I$. However, there are unique maps $I \rightarrow I$ and $I' \rightarrow I'$, namely id_I and $\text{id}_{I'}$. Thus, we must have $g \circ f = \text{id}_I$ and $f \circ g = \text{id}_{I'}$, so f and g are isomorphisms. ■

This allows us to refer to “the” initial or final object of a category, as it is essentially unique (if it exists).

Definition 1.2.7. If $A, B \in \text{ob } \mathcal{C}$, their product $A \times B$ is the final object in $\mathcal{C}_{A,B}$. That is, it is an object $A \times B$ together with maps $\pi_A : A \times B \rightarrow A$ and $\pi_B : A \times B \rightarrow B$ such that, whenever $Z \in \text{ob } \mathcal{C}$ with $f_A : Z \rightarrow A$ and $f_B : Z \rightarrow B$, there exists a unique $h : Z \rightarrow A \times B$ such that

$$\begin{array}{ccccc} & & & A & \\ & & f_A \nearrow & \nearrow \pi_A & \\ Z & \xrightarrow{\exists! h} & A \times B & & \\ & & f_B \searrow & \searrow \pi_B & \\ & & & B & \end{array}$$

commutes. Their coproduct $A \amalg B$ is the initial object in $\mathcal{C}^{A,B}$. That is, it is an object $A \amalg B$ together with maps $\iota_A : A \rightarrow A \amalg B$ and $\iota_B : B \rightarrow A \amalg B$ such that, for any $Z \in \text{ob } \mathcal{C}$ with $f_A : A \rightarrow Z$ and $f_B : B \rightarrow Z$, there exists a unique $h : A \amalg B \rightarrow Z$ such that

$$\begin{array}{ccccc} A & & f_A \searrow & & \\ & \searrow \iota_A & & \searrow \exists! h & \\ & & A \amalg B & \xrightarrow{\exists! h} & Z \\ & \nearrow \iota_B & & \nearrow f_B & \\ B & & & & \end{array}$$

commutes.

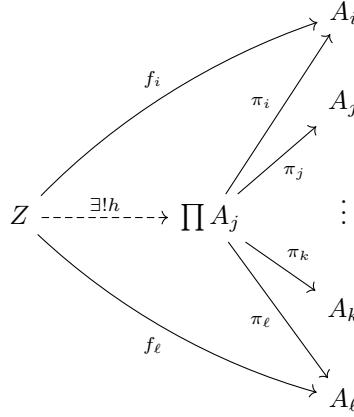
Just as initial and final objects need not exist in an arbitrary category, products and coproducts need not exist either. However, if they exist, they are unique up to isomorphism.

Example. The product in **Set** is the cartesian product $A \times B$ with the projection maps $\pi_A(a, b) = a$ and $\pi_B(a, b) = b$. If $f_A : Z \rightarrow A$ and $f_B : Z \rightarrow B$, the map h is given by $z \mapsto (f_A(z), f_B(z))$. The coproduct in **Set** is the disjoint union $A \sqcup B$ with the inclusion maps $\iota_A : a \mapsto a$ and $\iota_B : b \mapsto b$.

Example. The product between $a, b \in \text{ob Div}$ is $\text{gcd}(a, b)$, and their coproduct is $\text{lcm}(a, b)$.

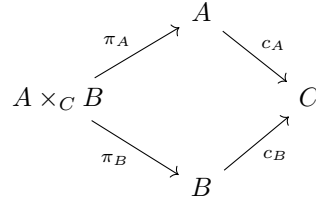
We can extend to a (co)product between more than two objects. If we define the category $\mathcal{C}_{A_i}$ as objects $(Z, f_i : Z \rightarrow A_i)$ and morphisms $(Z_1, f_i) \rightarrow (Z_2, g_i)$ consisting of morphisms $h : Z_1 \rightarrow Z_2$ such that $g_i \circ h = f_i$, then the product is the final object in $\mathcal{C}_{A_i}$. That is, $\prod A_j$ is equipped with maps $\pi_i : \prod A_j \rightarrow A_i$ such that, whenever

there are maps $f_i : Z \rightarrow A_i$, there exists a unique map $h : Z \rightarrow \prod A_j$ such that

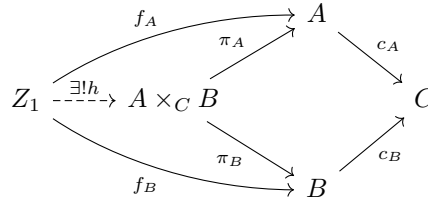


commutes. Similarly, the coproduct $\coprod A_j$ is the initial object in the category $\mathcal{C}^{A_i}$ that has objects $(Z, f_i : A_i \rightarrow Z)$ and morphisms $(Z_1, f_i) \rightarrow (Z_2, g_i)$ consisting of morphisms $h : Z_1 \rightarrow Z_2$ such that $h \circ f_i = g_i$.

If we have maps $c_A : A \rightarrow C$ and $c_B : B \rightarrow C$, we can define the *fibred* product as the final object in the fibred version of the category $\mathcal{C}_{A,B}$. That is, the fibred product $A \times_C B$ comes with maps $\pi_A : A \times_C B \rightarrow A$ and $\pi_B : A \times_C B \rightarrow B$ such that



commutes. Whenever Z also has such maps $f_A : Z \rightarrow A$ and $f_B : Z \rightarrow B$, then there exists a unique $h : Z \rightarrow A \times_C B$ such that



commutes. Similarly, we can define the fibred coproduct $A \amalg_C B$ as the initial object in the fibred version of $\mathcal{C}^{A,B}$. Just as with (co)products, we can also extend these to more than two objects. Fibred products are also sometimes called pullbacks, and fibred coproducts are sometimes called pushouts.

Example. The fibred product $A \times_C B$ in **Set** is $A \times B$ modulo the equivalence relation $a \sim b$ if $c_A(a) = c_B(b)$. The fibred coproduct $A \amalg_C B$ in **Set** is $A \sqcup B$ modulo the equivalence relation $c_A(c) \sim c_B(c)$ for all $c \in C$.

Example. The union of A and B is given by $A \cup B = A \amalg_{A \cap B} B$.

If F is final in $\mathcal{C}$, then every $A \in \text{ob } \mathcal{C}$ comes with a unique map $c_A : A \rightarrow F$. Additionally, if $f_A : Z \rightarrow A$ and $f_B : Z \rightarrow B$, we must have $c_A \circ f_A = c_B \circ f_B = c_Z$ by the uniqueness of maps into the final object. Thus, $A \times B$ satisfies the universal property of $A \times_F B$. Similarly, if I is initial, $A \amalg B = A \amalg_I B$. Thus, fibred (co)products are a more general version of regular (co)products (in the case when initial and final objects exist).

Fibred products and coproducts are a special case of *(co)limits*.

Definition 1.2.8. Let D be a diagram with objects D_i . The *limit* (inverse limit, projective limit) $\varprojlim D_i$ is the final object in $\text{cone}(D)$. The *colimit* (direct limit, inductive limit) $\varinjlim D_i$ is the initial object in $\text{cocone}(D)$.

Note that limits and colimits are the interesting universal objects in these categories: if I is initial and F is final in $\mathcal{C}$, then $(I, i : I \rightarrow D)$ is initial in $\text{cone}(D)$ (where i is the unique maps from I to D) and $(F, f : D \rightarrow F)$ is final in $\text{cocone}(D)$ (where f is the unique maps from D to F). Thus, passing to the (co)cone doesn't yield anything new in these situations.

The (co)product of A_i is the (co)limit of the diagram D consisting of A_i and only the identity maps between them. The fibered (co)product is the (co)limit of the diagram D consisting of A_i and C , together with the identity maps and the maps $c_A : A \rightarrow C$.

Another important limit is a *(co)equalizer*, which is the (co)limit of the diagram

$$A \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} B$$

Example. In **Set**, the equalizer is given by $\{a \in A : f(a) = g(a)\}$ and the coequalizer is given by B modulo the equivalence relation $f(a) \sim g(a)$ for all $a \in A$.

The constructions of (co)equalizers and fibered (co)products look quite similar, and there is good reason for this. While the equalizer is not quite the fibered product $A \times_B A$, we can construct an equivalent fibered product: the maps $f, g : A \rightarrow B$ correspond uniquely to a map $(f, g) : A \rightarrow B \times B$, and we also have a map $(\text{id}, \text{id}) : B \rightarrow B \times B$. The equalizer is the fibered product $A \times_{B \times B} B$ with these maps. Similarly, the coequalizer is the fibered coproduct $A \amalg_{A \amalg A} B$.

Equalizers are nice because they turn out to be precisely what we need to construct arbitrary limits.

Definition 1.2.9. A category is said to be *(co)complete* if it contains the (co)limit of all small diagrams.

The motivation for this terminology is by analogy with metric spaces, where a metric space is complete if all Cauchy sequences have a limit.

Theorem 1.2.10. *If a category has all (co)products and (co)equalizers, it is (co)complete.*

Proof. We'll prove completeness. Let $\mathcal{C}$ have all products and equalizers. Take a diagram D in $\mathcal{C}$, and index the objects by I (i.e. let $\mathcal{J}$ be a category with $\text{ob } \mathcal{J} = I$ and let D be a functor $\mathcal{J} \rightarrow \mathcal{C}$). Construct a product X consisting of one copy of D_i for each $i \in I$ and a product Y consisting of one copy of D_j for each morphism $D_i \rightarrow D_j$

$$X = \prod_{i \in I} D_i \qquad Y = \prod_{D_i \rightarrow D_j} D_j.$$

Now, the projection maps $X \rightarrow D_i$ give a map $X \rightarrow Y$, which we'll call id . For each factor D_j of Y corresponding to the map $D_i \rightarrow D_j$, we get a map $X \rightarrow D_i \rightarrow D_j$, and so this gives rise to another map $f : X \rightarrow Y$. We claim that the equalizer of

$$X \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{\text{id}} \end{array} Y$$

is the limit of the diagram D . Indeed, commuting with the maps of D is precisely the same as commuting with the diagram above, and so taking an equalizer is the same as taking a limit of D . ■

Corollary 1.2.11. *If a category contains all fibered (co)products, it is (co)complete.*

Proof. The category must contain the empty fibered (co)product, which is a final (initial) object. Thus, the category also contains all (co)products. We have already seen that (co)equalizers are fibered (co)products. ■

1.2.4 Functors and Natural Transformations

Definition 1.2.12. If $\mathcal{C}$ and $\mathcal{D}$ are categories, a *(covariant) functor* $F : \mathcal{C} \rightarrow \mathcal{D}$ is an assignment $F(A) \in \text{ob } \mathcal{D}$ for each $A \in \text{ob } \mathcal{C}$ and $F(f) \in \text{Hom}_{\mathcal{D}}(F(A), F(B))$ for each $f \in \text{Hom}_{\mathcal{C}}(A, B)$ such that $F(\text{id}_A) = \text{id}_{F(A)}$ and $F(g \circ f) = F(g) \circ F(f)$.

Functors are essentially the “structure-preserving maps” between categories. For instance, if $f : A \rightarrow B$ is an isomorphism with inverse $g : B \rightarrow A$, then $F(f)$ is an isomorphism with inverse $F(g)$, as $F(g) \circ F(f) = F(g \circ f) = F(\text{id}_A) = \text{id}_{F(A)}$ and $F(f) \circ F(g) = F(f \circ g) = F(\text{id}_B) = \text{id}_{F(B)}$.

Example. For every category $\mathcal{C}$, there is an identity functor $\mathcal{C} \rightarrow \mathcal{C}$ that sends each object and morphism to itself.

Example. If $\mathcal{C}$ and $\mathcal{D}$ are categories and $A \in \text{ob } \mathcal{D}$, there is a constant functor $F_A : \mathcal{C} \rightarrow \mathcal{D}$ that sends every object to A and every morphism to id_A .

Example. We can construct a functor $\mathbf{Set} \rightarrow \mathbf{Set}$ by sending a set S to its powerset $\mathcal{P}(S)$ and sending a map $f : S \rightarrow T$ to a map $\mathcal{P}(S) \rightarrow \mathcal{P}(T)$ that sends $A \subseteq S$ to $f(A) \subseteq T$.

Example. Functors give a more precise way of thinking about diagrams than just “objects and morphisms between them”. We’ll say a diagram of type $\mathcal{I}$ in category $\mathcal{C}$ is a functor $I : \mathcal{I} \rightarrow \mathcal{C}$. The categories $\mathcal{I}$ that we consider are usually categories stemming from posets. We may make the functors explicit by writing $\varprojlim_I D_i$ and $\varinjlim_I D_i$, where $D_i = I(i)$ for $i \in \text{ob } \mathcal{I}$.

The way we obtain $\mathcal{C}^{op}$ from $\mathcal{C}$ is similar to a functor in that it assigns an object of $\mathcal{C}^{op}$ to an object of $\mathcal{C}$ and a morphism of $\mathcal{C}^{op}$ to a morphism of $\mathcal{C}$. However, this assignment is orientation reversing, and gives rise to the notion of contravariant functors.

Definition 1.2.13. If $\mathcal{C}$ and $\mathcal{D}$ are categories, a *contravariant functor* $F : \mathcal{C} \rightarrow \mathcal{D}$ is an assignment $F(A) \in \text{ob } \mathcal{D}$ for each $A \in \text{ob } \mathcal{C}$ and $F(f) \in \text{Hom}_{\mathcal{D}}(F(B), F(A))$ for each $f \in \text{Hom}_{\mathcal{C}}(A, B)$ such that $F(\text{id}_A) = \text{id}_{F(A)}$ and $F(g \circ f) = F(f) \circ F(g)$.

Contravariant functors $\mathcal{C} \rightarrow \mathcal{D}$ are precisely the same as covariant functors $\mathcal{C}^{op} \rightarrow \mathcal{D}$.

Definition 1.2.14. A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is said to be:

- *Full* (or *locally surjective*) if the map $\text{Hom}_{\mathcal{C}}(A, B) \rightarrow \text{Hom}_{\mathcal{D}}(F(A), F(B))$ is surjective for all $A, B \in \text{ob } \mathcal{C}$
- *Faithful* (or *locally injective*) if the map $\text{Hom}_{\mathcal{C}}(A, B) \rightarrow \text{Hom}_{\mathcal{D}}(F(A), F(B))$ is injective for all $A, B \in \text{ob } \mathcal{C}$
- *Fully faithful* if the map $\text{Hom}_{\mathcal{C}}(A, B) \rightarrow \text{Hom}_{\mathcal{D}}(F(A), F(B))$ is bijective for all $A, B \in \text{ob } \mathcal{C}$.

Note that these are “local” rather than “global” conditions, in that these functors need not satisfy any conditions about how they act on objects. For instance, if $\mathcal{C}$ is the category

$$\text{id}_A \circlearrowleft A \xrightarrow{f} B \circlearrowright \text{id}_B$$

and $\mathcal{D}$ is the category

$$* \circlearrowright \text{id}_*$$

then there is a fully faithful functor $F : \mathcal{C} \rightarrow \mathcal{D}$ that sends A and B to $*$ and every morphism to id_* . However, it is not an “isomorphism of categories”: for instance, it has no left inverse.

Just as we can consider maps between categories, we can consider maps between functors:

Definition 1.2.15. If $F, G : \mathcal{C} \rightarrow \mathcal{D}$ are functors, a natural transformation α from F to G is a collection of maps $\alpha_A \in \text{Hom}_{\mathcal{D}}(F(A), G(A))$ for each $A \in \text{ob } \mathcal{C}$ such that

$$\begin{array}{ccc} F(A) & \xrightarrow{F(f)} & F(B) \\ \alpha_A \downarrow & & \downarrow \alpha_B \\ G(A) & \xrightarrow{G(f)} & G(B) \end{array}$$

commutes for every $A, B \in \text{ob } \mathcal{C}$ and $f \in \text{Hom}_{\mathcal{C}}(A, B)$.

Example. If $F_A : \mathcal{D} \rightarrow \mathcal{C}$ is the constant functor and $I : \mathcal{D} \rightarrow \mathcal{C}$ is some diagram, then a natural transformation $\alpha : F_A \rightarrow I$ satisfies the commutative diagram

$$\begin{array}{ccc} F_A(X) = A & \xrightarrow{F_A(f)} & F_A(Y) = A \\ \alpha_X \downarrow & & \downarrow \alpha_Y \\ I(X) = D_X & \xrightarrow{D(f)} & I(Y) = D_Y \end{array}$$

which collapses to

$$\begin{array}{ccc} A & & \\ \alpha_X \downarrow & \searrow \alpha_Y & \\ D_X & \xrightarrow{D(f)} & D_Y \end{array}$$

Thus, natural transformations correspond to cones over D . Similarly, natural transformations from I to constant functors correspond to cocones over D .

Example. Any functor $F : \mathcal{C} \rightarrow \mathcal{D}$ has the identity natural transformation $\text{id}_F : F \rightarrow F$ given by $(\text{id}_F)_A = \text{id}_{F(A)}$.

If $F, G, H : \mathcal{C} \rightarrow \mathcal{D}$ are functors and $\alpha : F \rightarrow G$ and $\beta : G \rightarrow H$ are natural transformations, we can construct a natural transformation $\beta \circ \alpha : F \rightarrow H$ given by $(\beta \circ \alpha)_A = \beta_A \circ \alpha_A$. This is a natural transformation because the two smaller squares in the diagram

$$\begin{array}{ccc} F(A) & \xrightarrow{F(f)} & F(B) \\ \alpha_A \downarrow & & \downarrow \alpha_B \\ G(A) & \xrightarrow{G(f)} & G(B) \\ \beta_A \downarrow & & \downarrow \beta_B \\ H(A) & \xrightarrow{H(f)} & H(B) \end{array} \quad \begin{array}{c} \beta_A \circ \alpha_A \swarrow \quad \searrow \beta_B \circ \alpha_B \end{array}$$

commute, which means that the outer arrows also commute. This process is called *vertical composition*. With this, we can create the *functor category*:

Definition 1.2.16. If $\mathcal{C}$ and $\mathcal{D}$ are categories, the *functor category* $[\mathcal{C}, \mathcal{D}]$ (also denoted $\mathcal{D}^{\mathcal{C}}$) has objects that are functors from $\mathcal{C}$ to $\mathcal{D}$ and morphisms that are natural transformations.

We'll say that a natural transformation $\alpha : F \rightarrow G$ is a *natural isomorphism* if it is an isomorphism in the functor category: that is, if there is a natural transformation $\beta : G \rightarrow F$ such that $\beta \circ \alpha$ and $\alpha \circ \beta$ are the identity transformations on F and G respectively.

Proposition 1.2.17. A natural transformation $\alpha : F \rightarrow G$ is a natural isomorphism if and only if each of the component maps α_A are isomorphisms.

Proof. Suppose that $\alpha : F \rightarrow G$ is a natural isomorphism with inverse $\beta : G \rightarrow F$. Then $\beta_A \circ \alpha_A = (\text{id}_F)_A = \text{id}_{F(A)}$ and $\alpha_A \circ \beta_A = (\text{id}_G)_A = \text{id}_{G(A)}$, so α_A is an isomorphism.

Conversely, suppose that each α_A is an isomorphism, and define β by $\beta_A = \alpha_A^{-1}$. It suffices to show that this is a natural transformation, as the composition will result in the necessary identity transformations. We thus want to show that the maps from $G(A)$ to $F(B)$ in

$$\begin{array}{ccc} F(A) & \xrightarrow{F(f)} & F(B) \\ \alpha_A^{-1} \uparrow \downarrow \alpha_A & & \alpha_B \downarrow \uparrow \alpha_B^{-1} \\ G(A) & \xrightarrow{G(f)} & G(B) \end{array}$$

agree. This can be achieved via a diagram chase:

$$F(f) \circ \alpha_A^{-1} = \alpha_B^{-1} \circ \alpha_B \circ F(f) \circ \alpha_A^{-1} = \alpha_B^{-1} \circ G(f) \circ \alpha_A \circ \alpha_A^{-1} = \alpha_B^{-1} \circ G(f).$$

■

1.2.5 Adjoints

Adjoints provide a notion of universality for functors.

Definition 1.2.18. If $F : \mathcal{C} \rightarrow \mathcal{D}$ and $G : \mathcal{D} \rightarrow \mathcal{C}$ are functors, we say F and G are *adjoints* (or that F is a left adjoint to G and G is a right adjoint to F) if, for every $A \in \text{ob } \mathcal{C}$ and $B \in \text{ob } \mathcal{D}$, there is a “natural” bijection $\tau_{A,B} : \text{Hom}_{\mathcal{D}}(F(A), B) \rightarrow \text{Hom}_{\mathcal{C}}(A, G(B))$ such that the diagram

$$\begin{array}{ccc} \text{Hom}_{\mathcal{D}}(F(A_1), B_1) & \xrightarrow{\tau_{A_1, B_1}} & \text{Hom}_{\mathcal{C}}(A_1, G(B_1)) \\ \downarrow g \circ \dots \circ F(f) & & \downarrow G(g) \circ \dots \circ f \\ \text{Hom}_{\mathcal{D}}(F(A_2), B_2) & \xrightarrow{\tau_{A_2, B_2}} & \text{Hom}_{\mathcal{C}}(A_2, G(B_2)) \end{array}$$

commutes for every $f : A_2 \rightarrow A_1$ and $g : B_1 \rightarrow B_2$. We'll write $\tau^{A,B} = \tau_{A,B}^{-1}$.

Example. There is a functor $F : \mathcal{C} \rightarrow [\mathcal{D}, \mathcal{C}]$ that sends each object A to the constant functor F_A (which sends objects to A and morphisms to id_A). If $I : \mathcal{D} \rightarrow \mathcal{C}$ is a diagram, the natural transformations (i.e. elements of $\text{Hom}_{[\mathcal{D}, \mathcal{C}]}(F_A, I)$) correspond to cones over D . Meanwhile, if $L(I) = \varprojlim_I D_i$, then elements of $\text{Hom}_{\mathcal{C}}(A, L(I))$ correspond naturally to cones over D by composing with the maps from $L(I)$ to D (this represents every cone uniquely because of the universality of limits). Thus, the limit is the right adjoint of the constant functor. It is possible to check the commutativity of the diagram, but we in general won't do this because it is more tedious than enlightening, so we'll accept natural-seeming bijections as sufficient proof of adjointness.

Example. For any set X , we have a functor $\cdot \times X : \mathbf{Set} \rightarrow \mathbf{Set}$ that sends a set A to $A \times X$ and a map $f : A \rightarrow B$ to $f \times \text{id} : A \times X \rightarrow B \times X$. We also have a functor $\text{Hom}(X, \cdot)$ that sends a set A to $\text{Hom}(X, A)$ and a map $f : A \rightarrow B$ to the postcomposition map $g \mapsto f \circ g$. We claim that $\cdot \times X$ and $\text{Hom}(X, \cdot)$ are adjoints. For any sets A and B , the desired bijection is given by sending $f \in \text{Hom}(A \times X, B)$ to $a \mapsto f(a, \cdot) \in \text{Hom}(X, B)$.

Proposition 1.2.19. *If G and G' are both right adjoints to F , then they are naturally isomorphic. If F and F' are both left adjoints to G , then they are naturally isomorphic.*

Proof. We show the first statement. Let G and G' both be right adjoints to F . We have

$$\text{Hom}_{\mathcal{C}}(A, G(B)) \xleftarrow{\tau_{A,B}} \text{Hom}_{\mathcal{D}}(F(A), B) \xrightarrow{\tau'_{A,B}} \text{Hom}_{\mathcal{C}}(A, G'(B))$$

For $B \in \text{ob } \mathcal{D}$, take $A = G(B)$ and define $\alpha_B : G(B) \rightarrow G'(B)$ as $\tau'_{G(B),B} \circ \tau^{G(B),B}(\text{id}_{G(B)})$. We claim that α is a natural transformation: that is, that

$$\begin{array}{ccc} G(B_1) & \xrightarrow{G(g)} & G(B_2) \\ \alpha_{B_1} \downarrow & & \downarrow \alpha_{B_2} \\ G'(B_1) & \xrightarrow{G'(g)} & G'(B_2) \end{array}$$

commutes for all $g : B_1 \rightarrow B_2$. The diagram

$$\begin{array}{ccccc} \text{Hom}_{\mathcal{C}}(G(B_1), G(B_1)) & \xleftarrow{\tau^{G(B_1),B_1}} & \text{Hom}_{\mathcal{D}}(FG(B_1), B_1) & \xrightarrow{\tau'_{G(B_1),B_1}} & \text{Hom}_{\mathcal{C}}(G(B_1), G'(B_1)) \\ \downarrow G(g) \circ \dots & \xleftarrow{\tau^{G(B_1),B_1}} & \downarrow g \circ \dots & & \downarrow G'(g) \circ \dots \\ \text{Hom}_{\mathcal{C}}(G(B_1), G(B_2)) & \xleftarrow{\tau^{G(B_1),B_2}} & \text{Hom}_{\mathcal{D}}(FG(B_1), B_2) & \xrightarrow{\tau'_{G(B_1),B_2}} & \text{Hom}_{\mathcal{C}}(G(B_1), G'(B_2)) \\ \dots \circ G(g) \uparrow & \xleftarrow{\tau^{G(B_1),B_2}} & \dots \circ FG(g) \uparrow & & \dots \circ G(g) \uparrow \\ \text{Hom}_{\mathcal{C}}(G(B_2), G(B_2)) & \xleftarrow{\tau^{G(B_2),B_2}} & \text{Hom}_{\mathcal{D}}(FG(B_2), B_2) & \xrightarrow{\tau'_{G(B_2),B_2}} & \text{Hom}_{\mathcal{C}}(G(B_2), G'(B_2)) \end{array}$$

commutes. Following $\text{id}_{G(B_1)}$ around the diagram gives

$$G'(g) \circ \alpha_{B_1} = G'(g) \circ (\tau'_{G(B_1),B_1} \circ \tau^{G(B_1),B_1})(\text{id}_{G(B_1)}) = \tau_{G(B_1),B_2}^{-1} \circ \tau'_{G(B_1),B_2}(G(g) \circ \text{id}_{G(B_1)}) = \tau^{G(B_1),B_2} \circ \tau'_{G(B_1),B_2}(G(g)),$$

and following $\text{id}_{G(B_2)}$ gives

$$\alpha_{B_2} \circ G(g) = (\tau'_{G(B_2),B_2} \circ \tau^{G(B_2),B_2})(\text{id}_{G(B_2)}) \circ G(g) = \tau^{G(B_1),B_2} \circ \tau'_{G(B_1),B_2}(\text{id}_{G(B_2)} \circ G(g)) = \tau^{G(B_1),B_2} \circ \tau'_{G(B_1),B_2}(G(g)).$$

Thus, $G'(g) \circ \alpha_{B_1} = \alpha_{B_2} \circ G(g)$, so α is indeed a natural transformation.

We now verify that each α_B is an isomorphism, so that α is a natural isomorphism. Define $\beta_B : G'(B) \rightarrow G(B)$ as $\tau_{G'(B),B} \circ \tau'^{-1}_{G'(B),B}(\text{id}_{G'(B)})$. We have the commuting diagram

$$\begin{array}{ccccc} \text{Hom}_{\mathcal{C}}(G(B), G(B)) & \xleftarrow{\tau^{G(B),B}} & \text{Hom}_{\mathcal{D}}(FG(B), B) & \xleftarrow{\tau'^{G(B),B}} & \text{Hom}_{\mathcal{C}}(G(B), G'(B)) \\ \downarrow \dots \circ \beta_B & \xleftarrow{\tau^{G(B),B}} & \downarrow \dots \circ F(\beta_B) & \xleftarrow{\tau'^{G(B),B}} & \downarrow \dots \circ \beta_B \\ \text{Hom}_{\mathcal{C}}(G'(B), G(B)) & \xleftarrow{\tau^{G'(B),B}} & \text{Hom}_{\mathcal{D}}(FG'(B), B) & \xleftarrow{\tau'^{G'(B),B}} & \text{Hom}_{\mathcal{C}}(G'(B), G'(B)) \end{array}$$

Starting with $\text{id}_{G(B)}$ and going right and down gives $\alpha_B \circ \beta_B$, while going down and right gives $\text{id}_{G'(B)}$, so they are equal. Similarly, $\beta_B \circ \alpha_B = \text{id}_{G(B)}$, so β_B is an inverse to α_B . Thus, α is indeed a natural isomorphism. ■

Adjoints are particularly nice because of the following property:

Proposition 1.2.20. *Right adjoints commute with limits, and left adjoints commute with colimits.*

Proof. We show the first statement. Let $F : \mathcal{C} \rightarrow \mathcal{D}$ and $G : \mathcal{D} \rightarrow \mathcal{C}$ be adjoints, and let $I : \mathcal{J} \rightarrow \mathcal{D}$ be a diagram. We'll show that $G(\varprojlim I)$ satisfies the universal property of the limit of the diagram GI .

Let $c : C \rightarrow GI$ (that is, let c be a collection of maps from $C \in \text{ob } \mathcal{C}$ to the diagram GI). There is then $\tau_{C,I}^{-1}(c) : F(C) \rightarrow I$, and the fact that F is a left adjoint means that these maps commute with the diagram morphisms of I . Thus, there's a unique map $f : F(C) \rightarrow \varprojlim I$ such that the respective maps to I commute. Now, $\tau_{F(C), \varprojlim I}(f) : C \rightarrow G(\varprojlim I)$ is a map that makes the respective maps to GI commute. Thus, $G(\varprojlim I)$ satisfies the desired universal properties. ■

Functors that preserve limits are called *continuous* (by analogy with metric spaces), and functors that preserve colimits are called *cocontinuous*. Thus, we see that right adjoints are continuous and left adjoints are cocontinuous.

Remark 1.2.21. Throughout algebra, we tend to work with “sets with extra structure”. There is a “forgetful functor” U that sends an algebraic structure to its underlying set. Often, there is a “free functor” that, when given a set, produces the “most general” algebraic structure possible, and serves as a left adjoint to U . Because of this, products of these algebraic structures will look very familiar (since their base set will just be the product in **Set**), while coproducts may be significantly more complicated. It is possible that U is also a left adjoint, such as in the category of topological spaces.

An adjoint can equivalently be thought of natural isomorphisms $\tau_{\cdot, B} : \text{Hom}_{\mathcal{D}}(F(\cdot), B) \rightarrow \text{Hom}_{\mathcal{C}}(\cdot, GB)$ and $\tau_{A, \cdot} : \text{Hom}_{\mathcal{D}}(F(A), \cdot) \rightarrow \text{Hom}_{\mathcal{C}}(A, G(\cdot))$. Adjoints playing nice with precomposition is precisely the same as $\tau_{\cdot, B}$ being a natural transformation, and adjoints playing nice with postcomposition is the same as $\tau_{A, \cdot}$ being a natural transformation.

For each object A , we have $\eta_A = \tau_{A, FA}(\text{id}_{FA})$ as a distinguished element of $\text{Hom}_{\mathcal{C}}(A, GFA) \cong \text{Hom}_{\mathcal{D}}(FA, FA)$. This defines a natural transformation $\eta : \text{id}_{\mathcal{C}} \rightarrow GF$, called the *unit* of the adjunction. Similarly, $\epsilon_B = \tau^{GB, B}(\text{id}_{GB})$ is a distinguished element of $\text{Hom}_{\mathcal{D}}(FGB, B) \cong \text{Hom}_{\mathcal{C}}(GB, GB)$, giving a natural transformation $\epsilon : FG \rightarrow \text{id}_{\mathcal{D}}$ called the *counit* of the adjunction.

The unit and counit completely determine the bijection τ : if $f : F(A) \rightarrow B$, then

$$\tau_{A, B}(f) = \tau_{A, B}(f \circ \text{id}_{F(A)}) = Gf \circ \tau_{A, F(A)}(\text{id}_{F(A)}) = Gf \circ \eta_A.$$

Similarly, if $g : A \rightarrow G(B)$, then $\tau^{A, B}(g) = \epsilon_B \circ Fg$. In fact, given any two natural transformations $\eta : \text{id}_{\mathcal{C}} \rightarrow GF$ and $\epsilon : FG \rightarrow \text{id}_{\mathcal{D}}$ satisfying $\epsilon_{F(A)} \circ F\eta_A = \text{id}_{F(A)}$ and $G\epsilon_B \circ \eta_{G(B)} = \text{id}_{G(B)}$, we can obtain an adjunction between F and G (these two relations are precisely what it means for the expressions of τ and τ^{-1} in terms of the unit and counit to be inverses).

Example. Let us investigate the unit and counit in the adjunction between $\cdot \times X$ and $\text{Hom}(X, \cdot)$. The identity map $Y \times X \rightarrow Y \times X$ becomes the map $y \mapsto (y, \cdot)$, where $(y, \cdot) : x \mapsto (y, x)$ is an element of $\text{Hom}(X, Y \times X)$. The identity map $\text{Hom}(X, Y) \rightarrow \text{Hom}(X, Y)$ becomes the evaluation map $(f, x) \mapsto f(x)$ from $\text{Hom}(X, Y) \times X \rightarrow Y$.

One last remark we make is that, if we have a morphism $g : A \rightarrow G(B)$, then there is a unique morphism $f : F(A) \rightarrow B$ making the following diagram commute

$$\begin{array}{ccc} A & \xrightarrow{g} & G(B) \\ \eta_A \downarrow & \nearrow Gf & \\ GF(A) & & \end{array}$$

The morphism f is given by $\tau^{A, B}(g)$. This is a “universal property” of the functor GF together with the natural transformation η (many of the universal properties we encounter will be precisely this statement with concrete choices of functors).